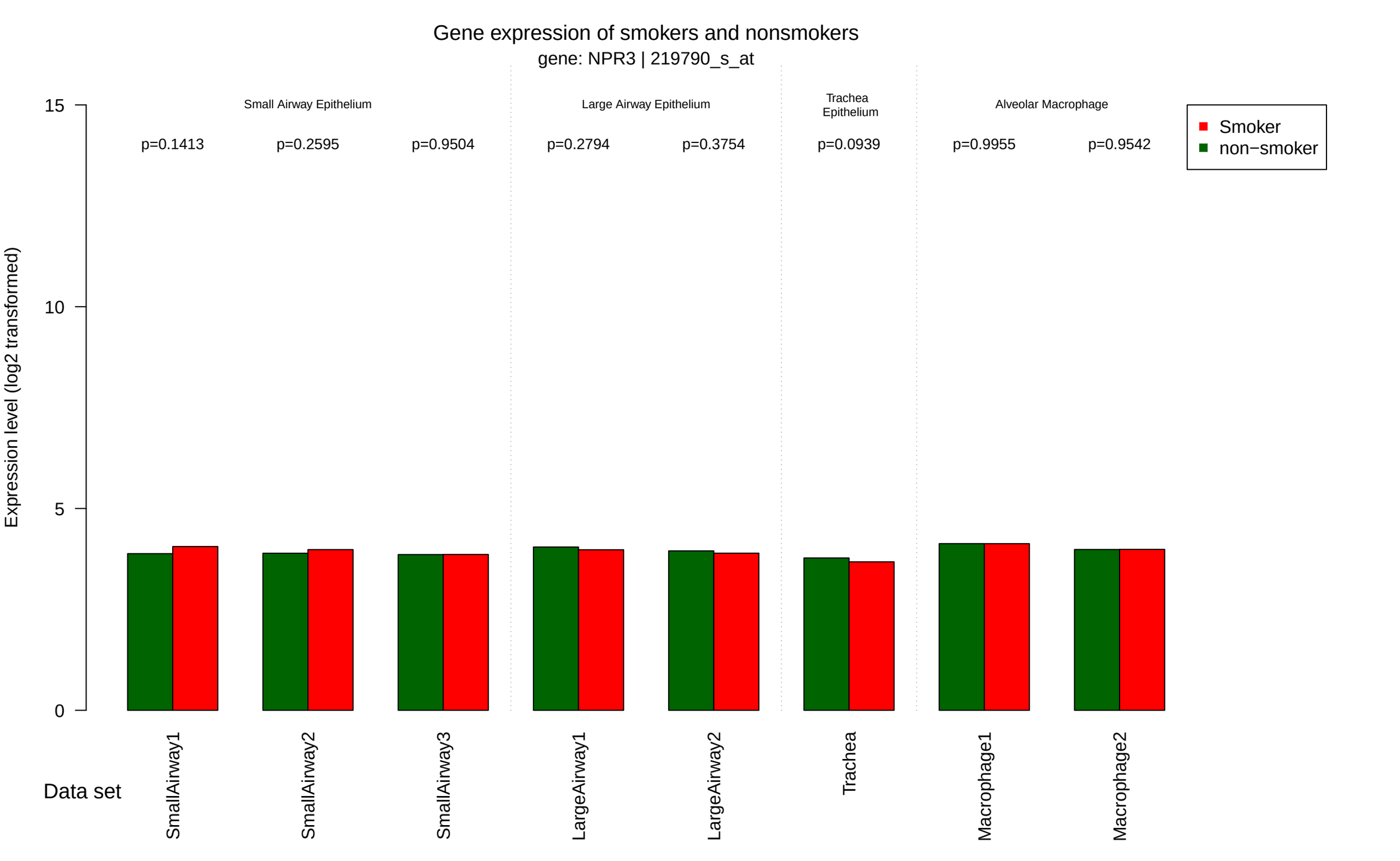
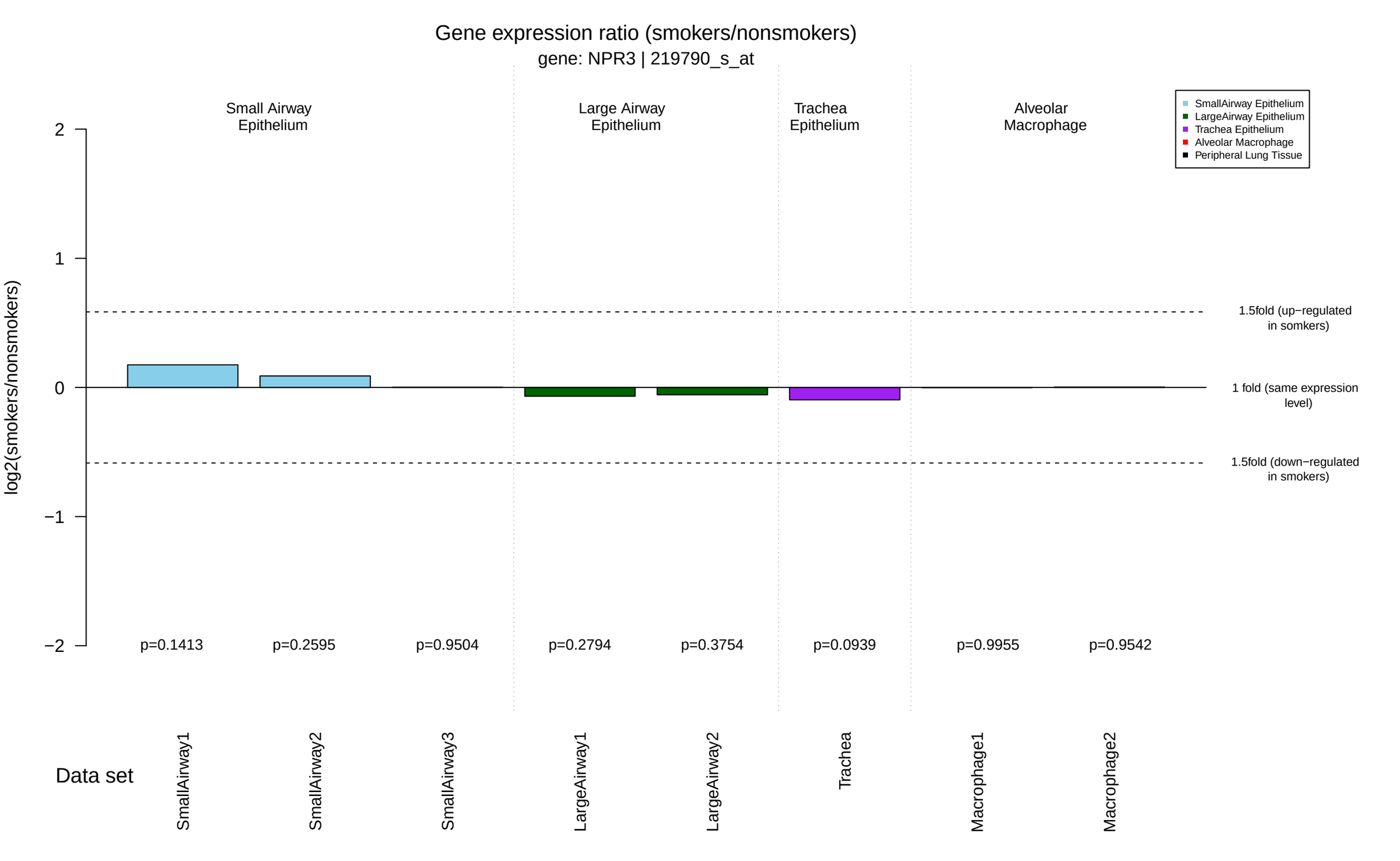
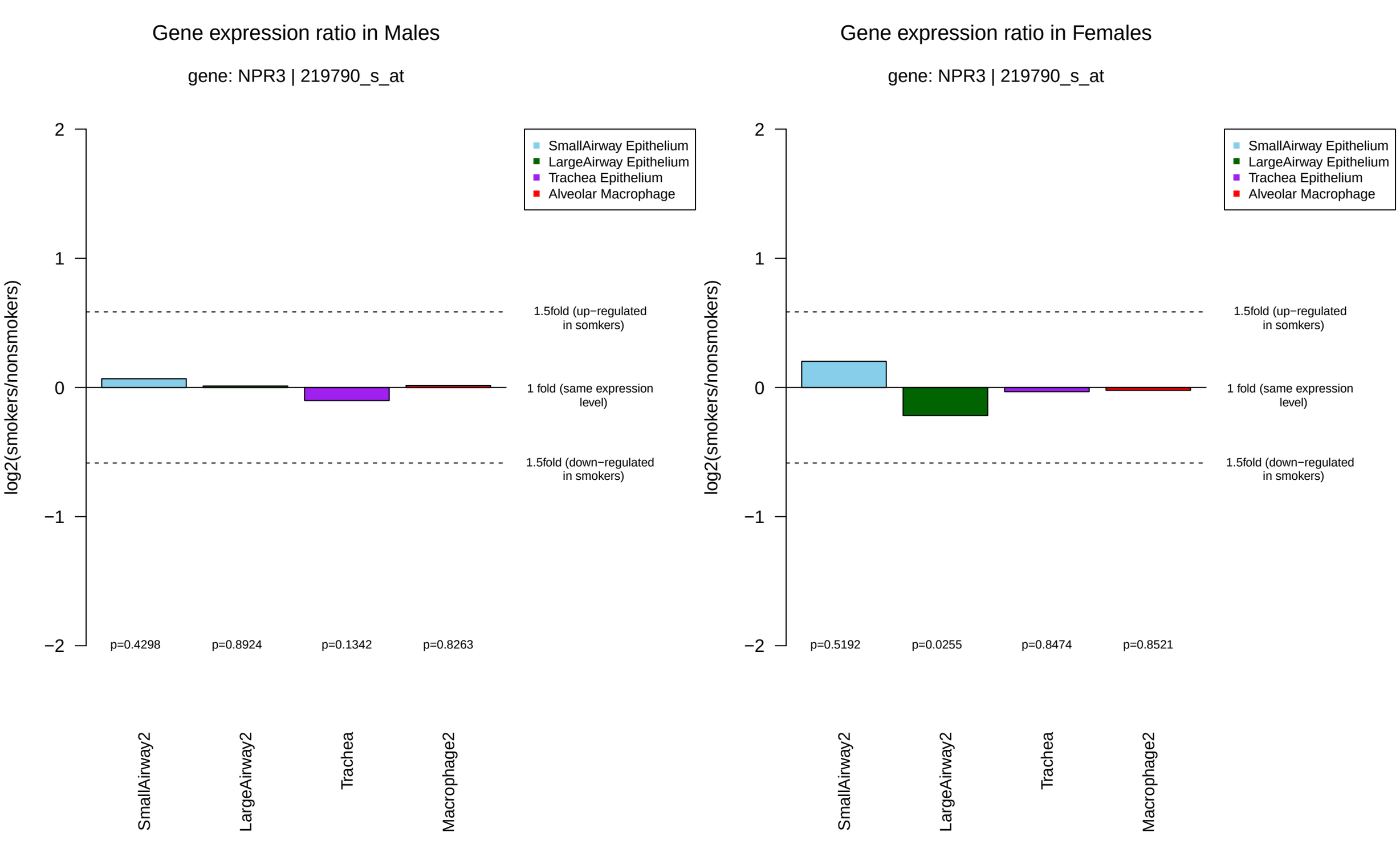
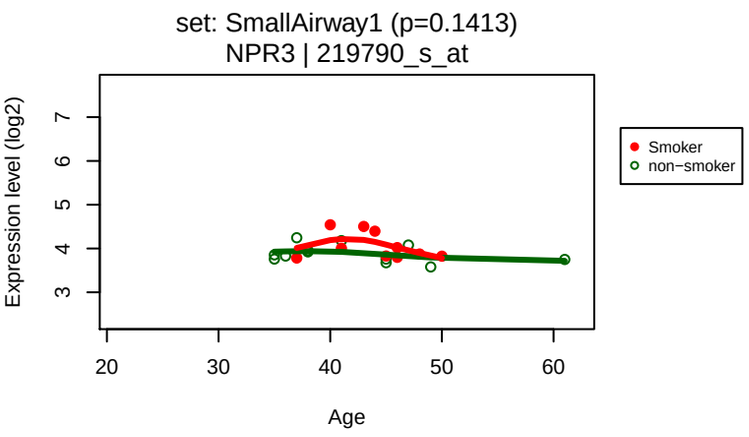
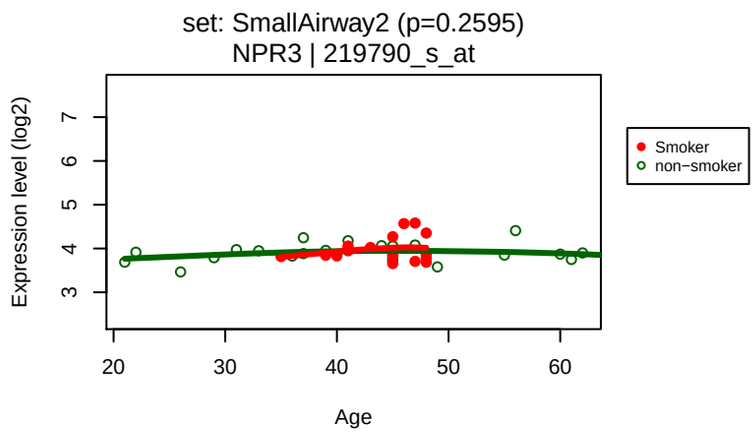
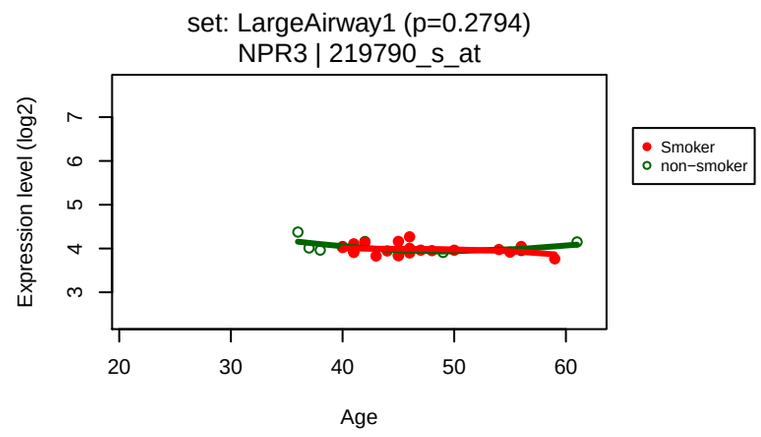

Gene expression of smokers and nonsmokers
gene: NPR3 | 219790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1413
p=0.2595
p=0.9504
p=0.2794
p=0.3754
p=0.0939
p=0.9955
p=0.9542
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

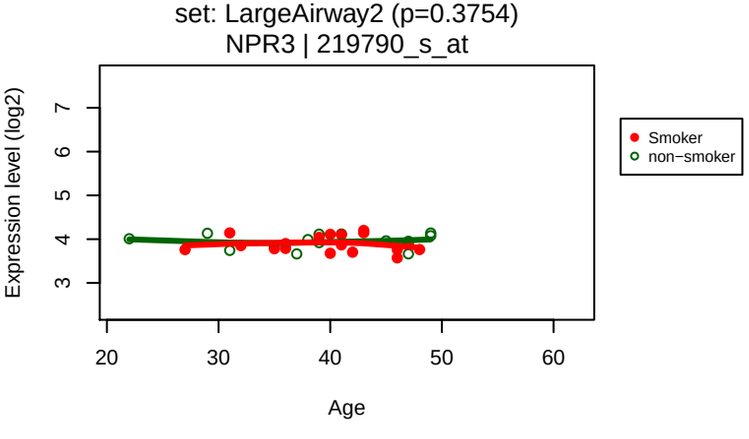

Gene expression ratio (smokers/nonsmokers)
gene: NPR3 | 219790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1413
p=0.2595
p=0.9504
p=0.2794
p=0.3754
p=0.0939
p=0.9955
p=0.9542
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

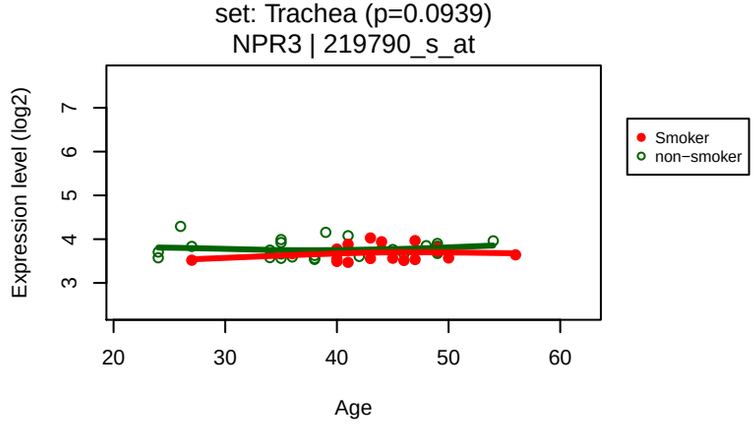

Gene expression ratio in Males
gene: NPR3 | 219790_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4298
p=0.8924
p=0.1342
p=0.8263
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPR3 | 219790_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5192
p=0.0255
p=0.8474
p=0.8521
SmallAirway2
LargeAirway2
Trachea
Macrophage2

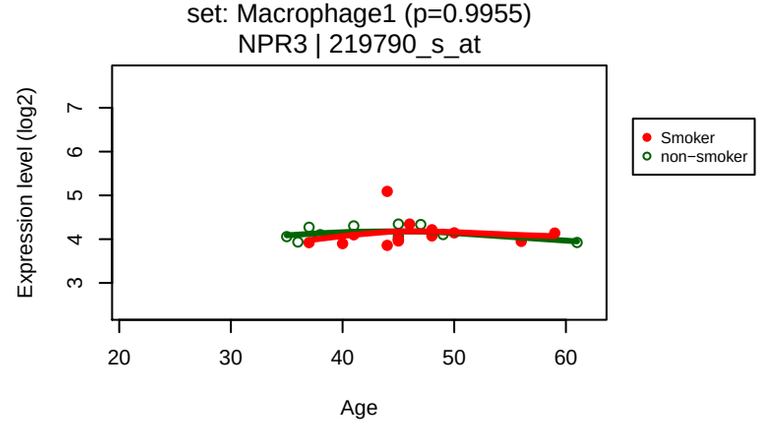

set: SmallAirway1 (p=0.1413)
NPR3 | 219790_s_at
Expression level (log2)
Age
Smoker
non−smoker

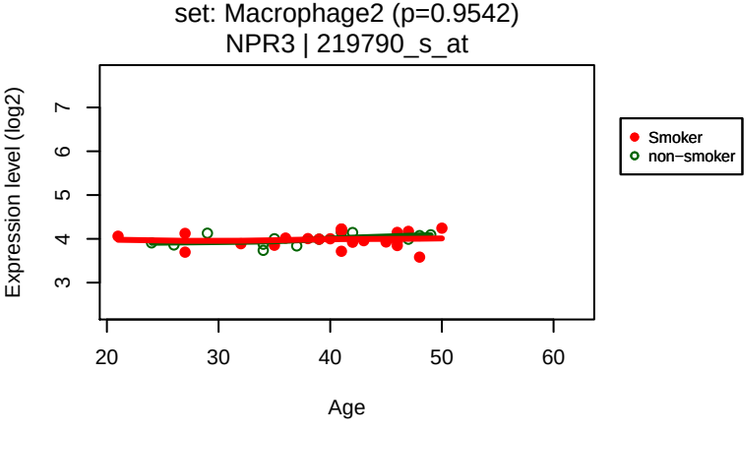

set: SmallAirway2 (p=0.2595)
NPR3 | 219790_s_at
Expression level (log2)
Age
Smoker
non−smoker

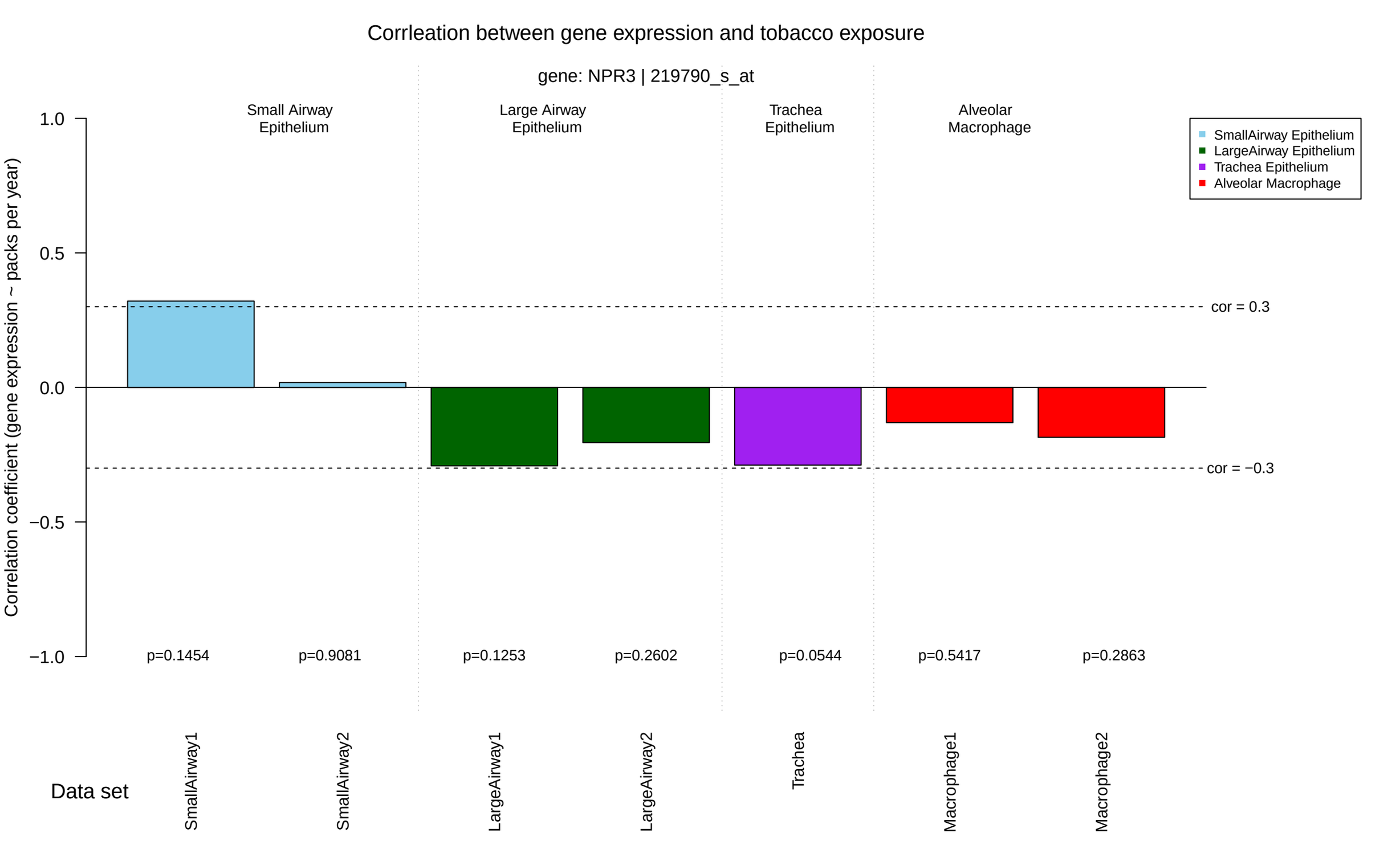

set: LargeAirway1 (p=0.2794)
NPR3 | 219790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3754)
NPR3 | 219790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0939)
NPR3 | 219790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9955)
NPR3 | 219790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9542)
NPR3 | 219790_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPR3 | 219790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1454
p=0.9081
p=0.1253
p=0.2602
p=0.0544
p=0.5417
p=0.2863
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2